

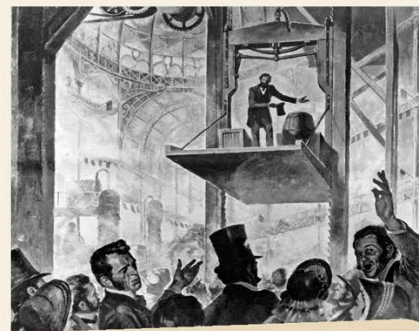


▲ SCRAPING THE SKY

The New York City skyline continues to rise, as workers put the finishing touches on the Empire State Building. This 102-story steel-frame skyscraper opened in 1931. It was much bigger than the world's first steel skyscraper, a 10-story building constructed in Chicago in 1885.

STEEL STRUCTURES

Steel is mostly iron combined with a little carbon—and is much stronger than pure iron. China and India had steel industries in ancient times, but this material took off in 1856 when Englishman Henry Bessemer invented a process to produce large amounts of it cheaply. This method was used to make ships, buildings, and armor up until the 1960s.



SAFETY ELEVATOR

American Elisha Otis demonstrated the first passenger safety elevator in the 1850s in New York City. This invention removed one of the major drawbacks of building high—too many stairs.



GOING UP!

The first escalator, invented by American engineer Jesse Reno, was just a sloping, moving walkway, but George Wheeler added folding steps. It was marketed by the Otis Elevator Company and by 1901 was appearing in some stores, as seen here in Boston.

THE CURTAIN WALL

Early steel-frame buildings still supported heavy stone or brick walls. In 1918, however, walls made of lightweight steel and glass that hung from the frame were introduced—these were called curtain walls.

The Bauhaus building in Dessau, Germany, uses curtain walls

